GRADE 2 · DEBUGGING & READING CODE

The Missing Block

A script lands short — add the missing block

Name: ______________________ Date: ______________________

★ **I can find a script that is missing a block and add the block it needs.**

Pixel stops too soon

Bit (green) and Pixel (purple) both start on the green flag. They should MEET on number 3. Bit reaches 3, but Pixel stops too soon — its script is MISSING a block. The dashed box shows where a block is missing. Read both, work out the missing move, and add it.

The goal: meet on 3

0	1	2	3	4	5
Bit			★ goal		Pixel

Bit's script (correct)

⚑ when green flag clicked

forward 3

Pixel's script (a block is missing)

⚑ when green flag clicked

back 1

[ ]

1 Run it as it is

Run Pixel with just its one move from home 5: back 1. What number does Pixel land on? Does it reach the goal 3?

____

____

2 What is missing?

Pixel is on 4 but the goal is 3. How many more squares back does Pixel need to go? What ONE block is missing?

____

3 Add the missing block

Write the missing block in the dashed box: ____ ____. After you add it, where does Pixel land? Do both meet on 3 now?

____

____

Big idea

Sometimes a script is missing a block, so it stops before the goal. Count how far the sprite still needs to go, then ADD the block that finishes the job. Here Pixel needed one more 'back 1' to reach 3.

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Quick facilitation notes & answer key

What this worksheet teaches

Students fix a missing-block bug by writing the move a set needs. The goal is to meet on 3; Bit is right, but Pixel has only one "back 1" (lands on 4) with an empty slot showing a block is missing. No coding experience needed.

Before you start (no computer needed)

No computer needed — paper and a pencil. You read the prompts aloud. Optional: two counters on a number line 0 to 5 (green Bit, purple Pixel) so children can trace both sets of moves and see which one misses the goal.

How to lead it (about 20 minutes)

1. Show them (you do). Pixel stops short of the goal — it needs one more move.
2. Do one together. Exercise 1 — Pixel's one move lands on 4 (one short of 3).
3. Let them try. Students work out the missing move and add it (Ex 2–3).

Answer key (these are the verified correct answers)

Exercise 1 — Pixel: 5, back 1 → 4. It does not reach the goal 3 (one short).

Exercise 2 — Pixel is on 4 and needs 3, so it needs one more square back; the missing block is back 1.

Exercise 3 — add back 1 → Pixel: 5 → 4 → 3. Both meet on 3. (A move that lands on 3 is correct; back 1 is simplest.)

If students get stuck — and ways to adjust

Watch for: changing the existing move instead of adding one. The fix is to add a step.

Make it easier: count how far Pixel still needs to go, then write that move.

Make it harder: ask for a single block (back 1) versus another way to reach 3.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential and concurrent events.

In plain terms: students write and run instructions, including two sets that run at the same time.

C3.2 — read and alter existing code, including code that involves sequential and concurrent events, and describe how changes to the code affect the outcomes.

In plain terms: students read two sets of instructions, find and fix the wrong one, and explain the fix.

You'll know it worked when

The child adds the missing move so Pixel reaches the goal.